HDOC Day-48

Aim: Write a C program using a pointer and a function named pivot_index() to find the leftmost index where the sum of elements to its left equals the sum of elements to its right. Print -1 if no such index exists.

C Program:

```
#include <stdio.h>

void pivot_index(int *arr, int n)
{
    int i, total = 0, left = 0, right;

    for (i = 0; i < n; i++)
        total += *(arr + i);

    for (i = 0; i < n; i++)
    {
        right = total - left - *(arr + i);
        if (left == right)
        {
            printf("Pivot index: %d\n", i);
            return;
        }
        left += *(arr + i);
    }
    printf("-1\n");
}

int main()
{
    int n, i;
    printf("Enter size of array: ");
    scanf("%d", &n);
    if (n <= 0)
    {
        printf("Invalid size\n");
        return 0;
    }
    int arr[n];
    printf("Enter array elements: ");
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    pivot_index(arr, n);
    return 0;
}
```

Sample Outputs:

Example 1

```
Input:
6
1 7 3 6 5 6
Output:
Pivot index: 3
```

Example 2

```
Input:
3
1 2 3
Output:
-1
```

Example 3

```
Input:
3
2 1 -1
Output:
Pivot index: 0
```